PROJECT 010

Hyperparameter Optimization

Mastering Grid, Random, and Bayesian Optimization for ML performance.

1 Project Overview

I tackled two regression problems using Grid Search, Random Search, and Bayesian Optimization (Optuna). My goal was to optimize hyperparameters for improved model performance, yielding an Rš of 0.9741 on the 3D Printer dataset and 0.9733 on the PLL Loop Filter dataset. These strategies showed distinct advantages depending on the search space.

Grid Search: Exhaustively tests every parameter combination.

Random Search: Samples parameter combinations randomly for exploration.

Bayesian Optimization: Utilizes surrogate models for efficient searching.

Optuna: State-of-the-art framework for hyperparameter optimization.

2 Datasets

2.1 3D Printer Extrusion Quality

Property	Value
Description	Dataset focuses on the quality of 3D prints based on operational parameters.
Total Rows	5,000
Columns	4 (3 features + 1 target)
Features	Temperature, Speed, Layer Height
Class Balance	Continuous target, 0-1 scale
Train / Test Split	80/20

2.2 PLL Loop Filter Tuning

Property	Value
Description	Dataset aimed at tuning phase-locked loop parameters post-silicon.
Total Rows	5,000
Columns	4 (3 features + 1 target)
Features	PLL Parameters
Class Balance	Continuous, microseconds
Train / Test Split	80/20

3 Methodology

3.1 Algorithms Used

Algorithm	Description
Grid Search	Chosen for exhaustive parameter testing.
Random Search	Ideal for quick explorations of larger spaces.
Bayesian Optimization	Utilizes past evaluations for intelligent sampling.
Optuna	Used for advanced Bayesian optimization and pruning.

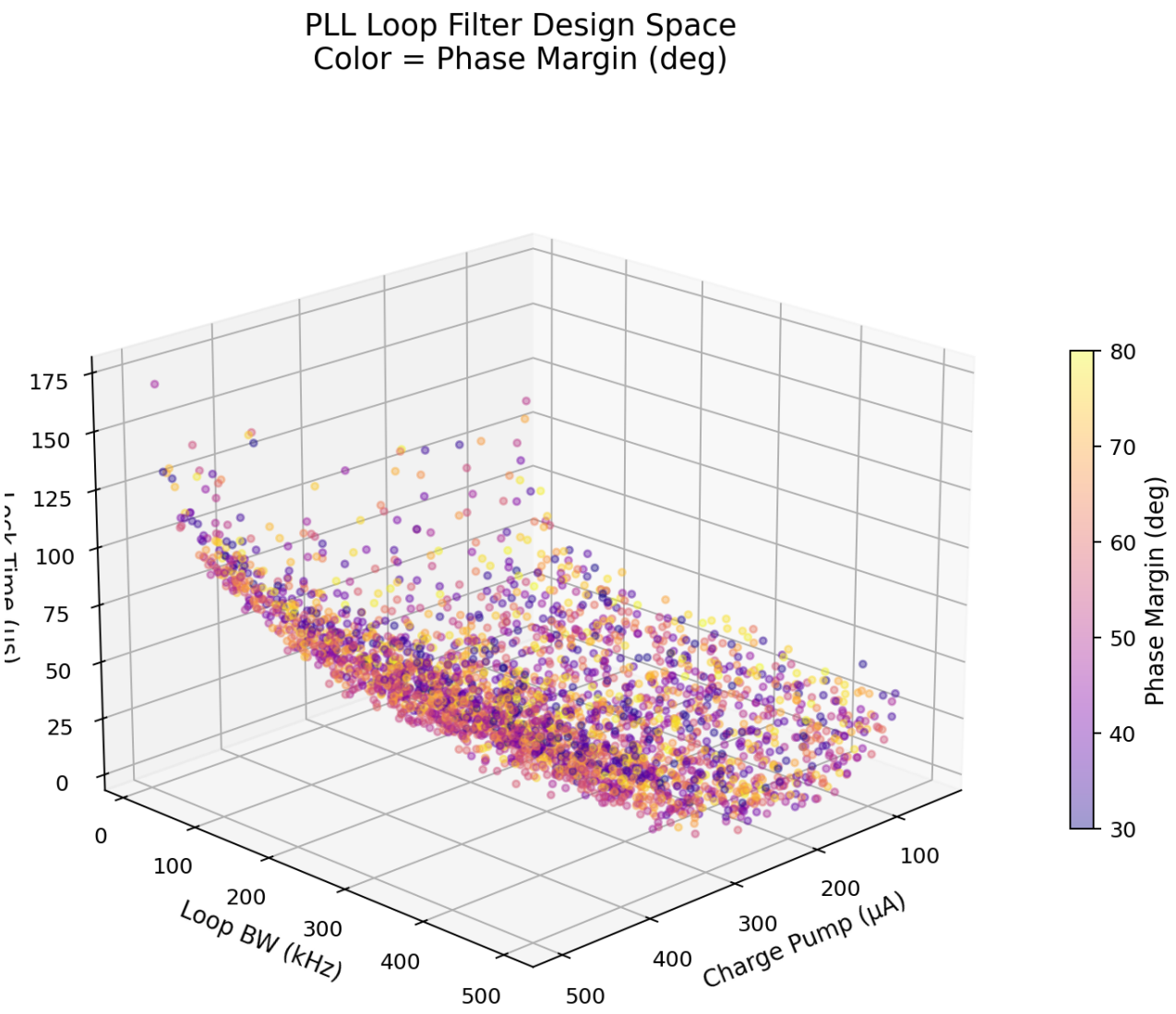
I loaded the data, established baselines, applied each optimization technique, and evaluated model performance using Rš and RMSE metrics.

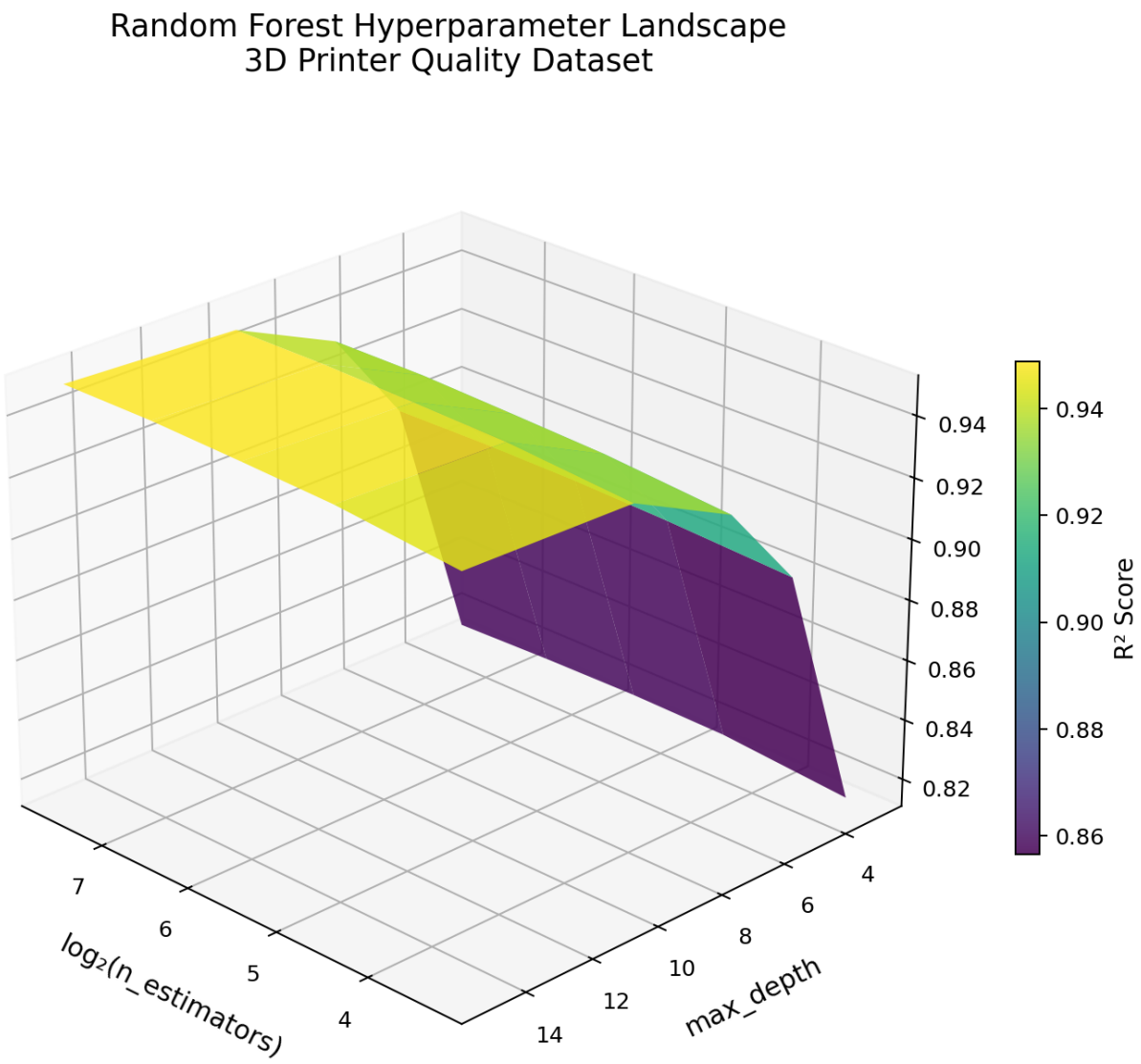
4 Results

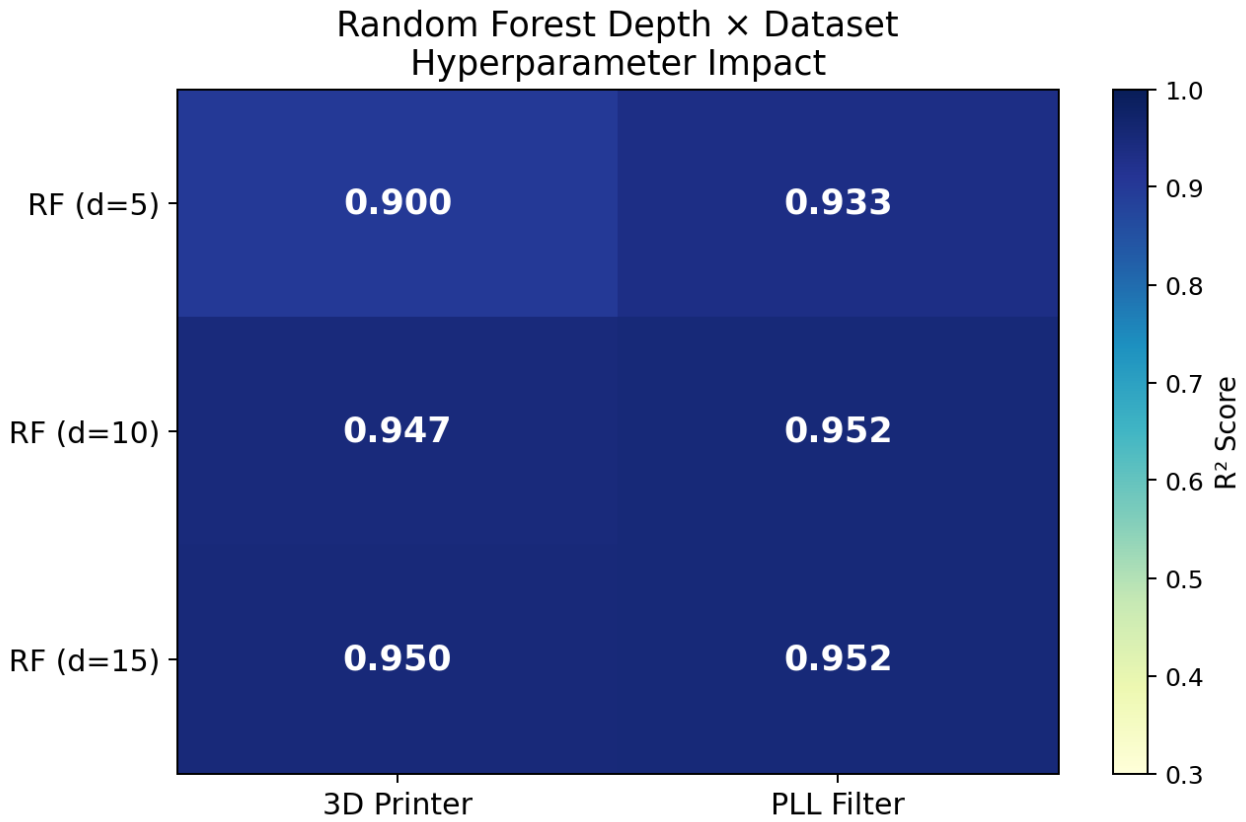
The Bayesian Optimization approach via Optuna outperformed others, significantly improving the output quality by efficiently navigating large parameter spaces. I found that random sampling provided satisfactory results but lacked the precision achieved with Bayesian methods.

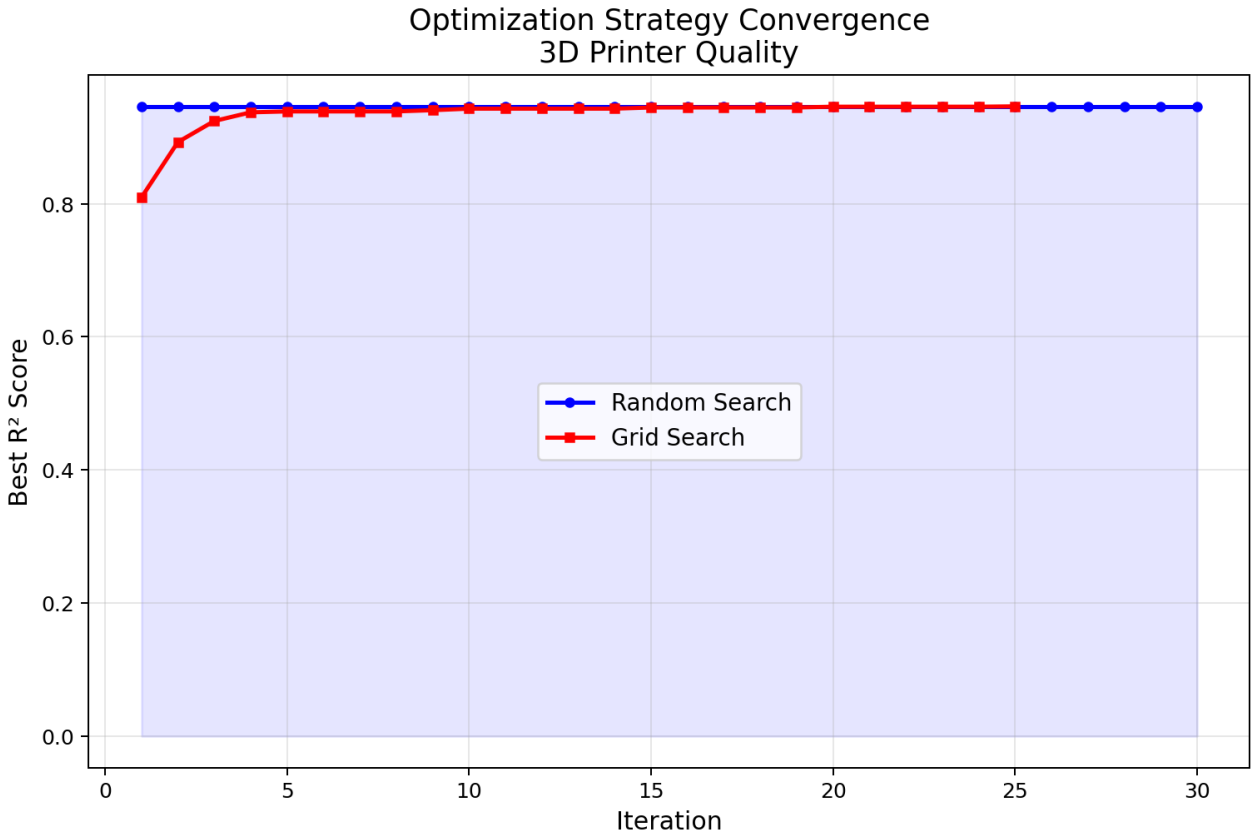
Metric	Value
Rš	0.9741
RMSE	0.0295
Rš	0.9733
RMSE	5.0692

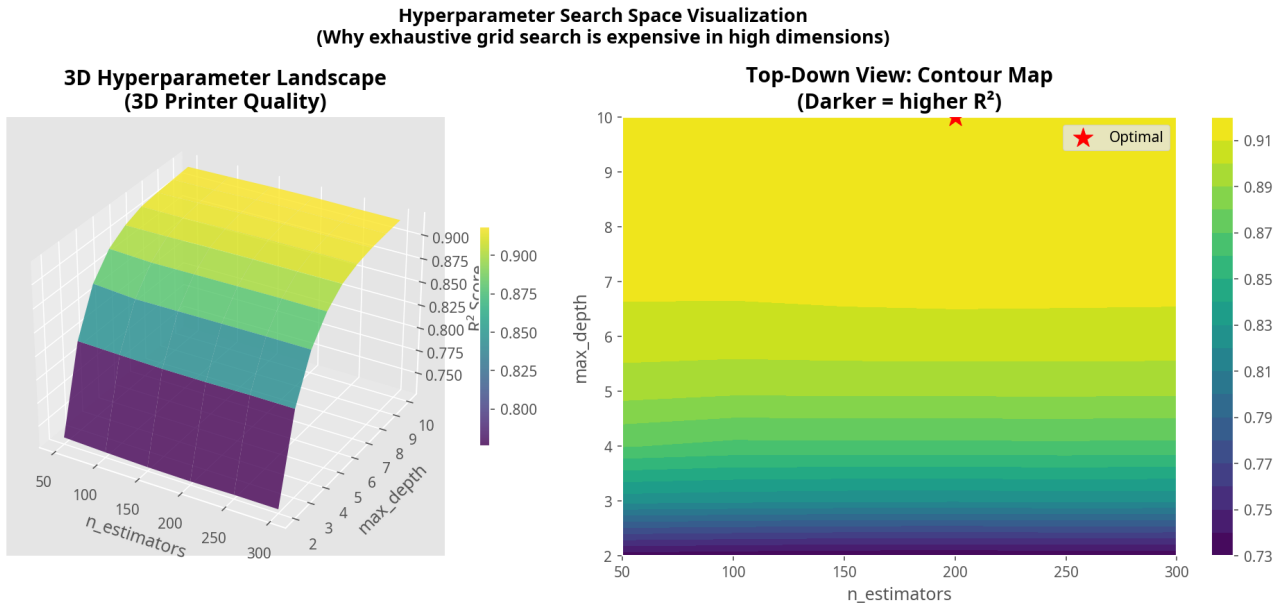
Best Result **Rš = 0.9741**

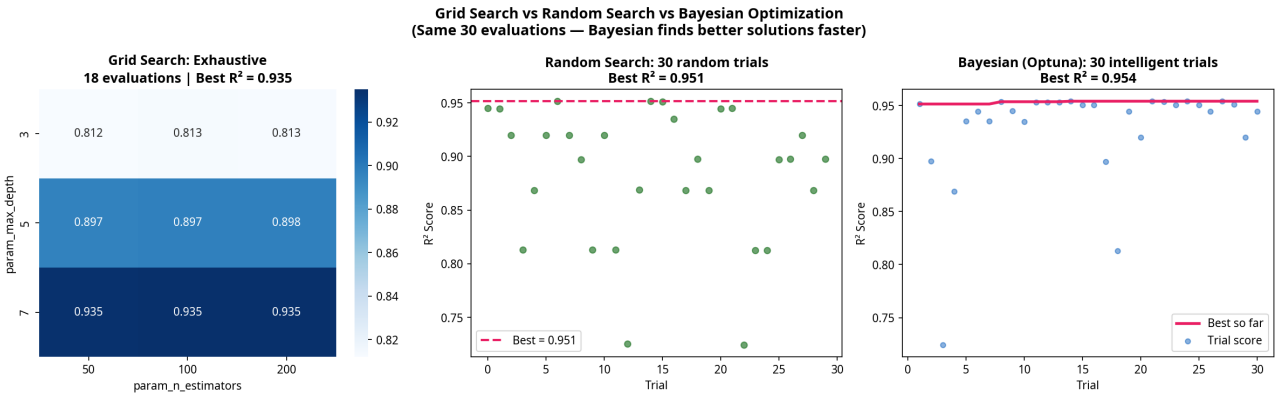

PLL Loop Filter Design Space
Color = Phase Margin (deg)
Lock Time (µs)
175
150
125
100
75
50
25
0
Loop BW (kHz)
0
100
200
300
400
500
Charge Pump (µA)
100
200
300
400
500
Phase Margin (deg)
80
70
60
50
40
30

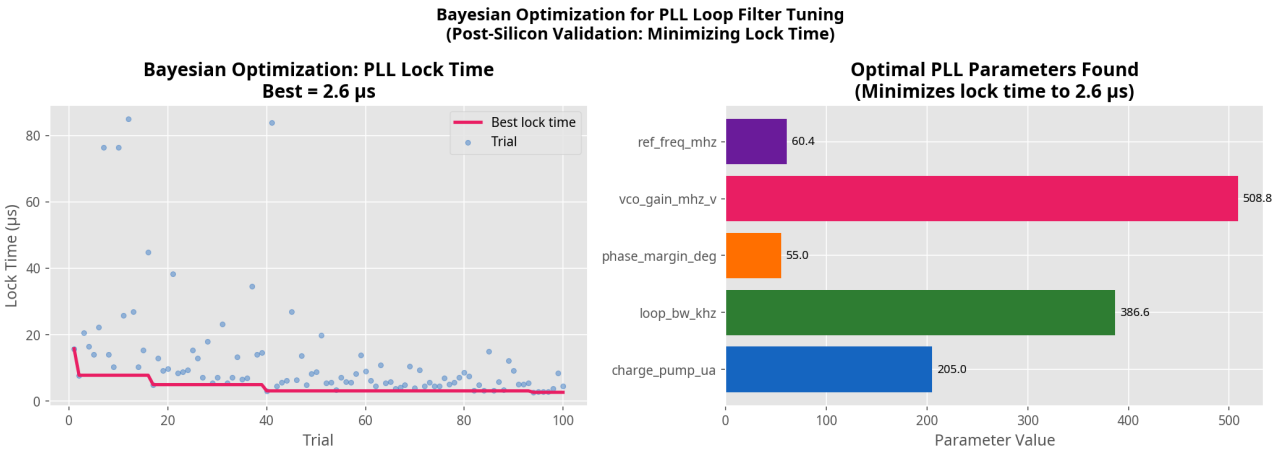

Random Forest Hyperparameter Landscape
3D Printer Quality Dataset
0.94
0.92
0.90
0.88
0.86
0.84
0.82
R² Score
0.94
0.92
0.90
0.88
0.86
7
6
5
4
log₂(n_estimators)
4
6
8
10
12
14
max_depth

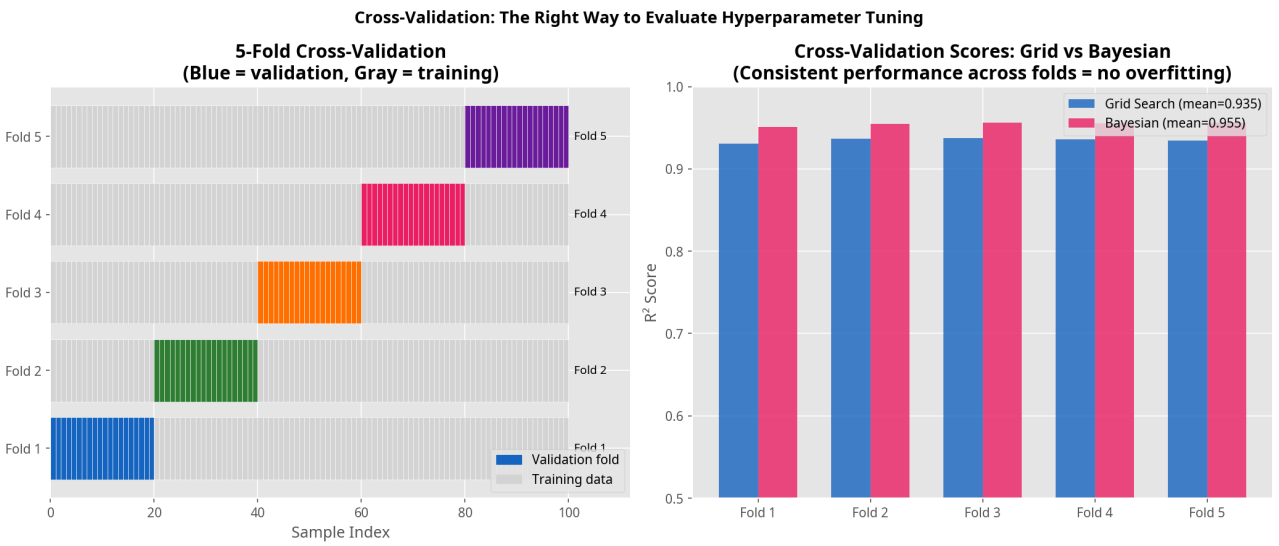

Random Forest Depth × Dataset
Hyperparameter Impact
RF (d=5)
RF (d=10)
RF (d=15)
0.900
0.933
0.947
0.952
0.950
0.952
3D Printer
PLL Filter
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
R² Score

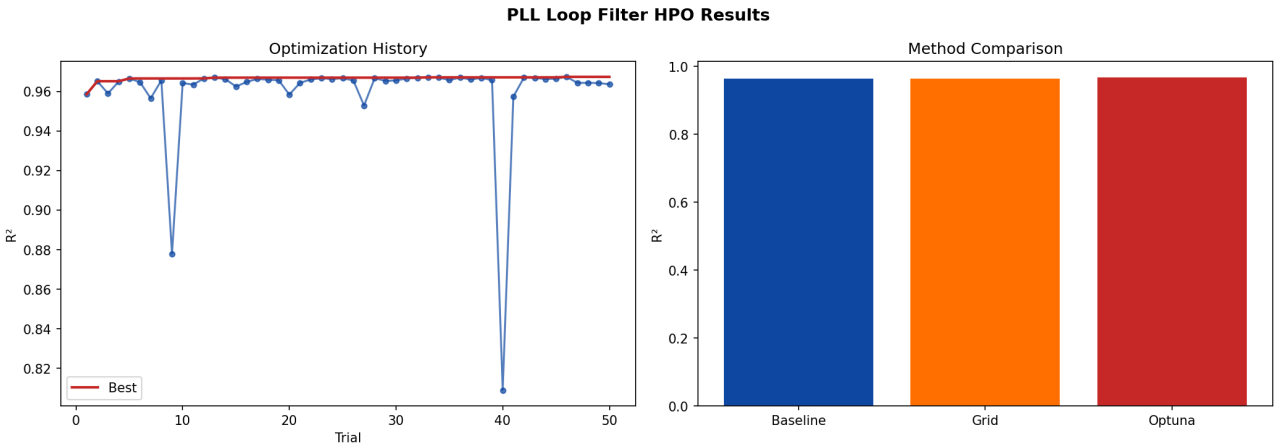

Optimization Strategy Convergence
3D Printer Quality
Best R² Score
0.0
0.2
0.4
0.6
0.8
0
5
10
15
20
25
30
Iteration
Random Search
Grid Search

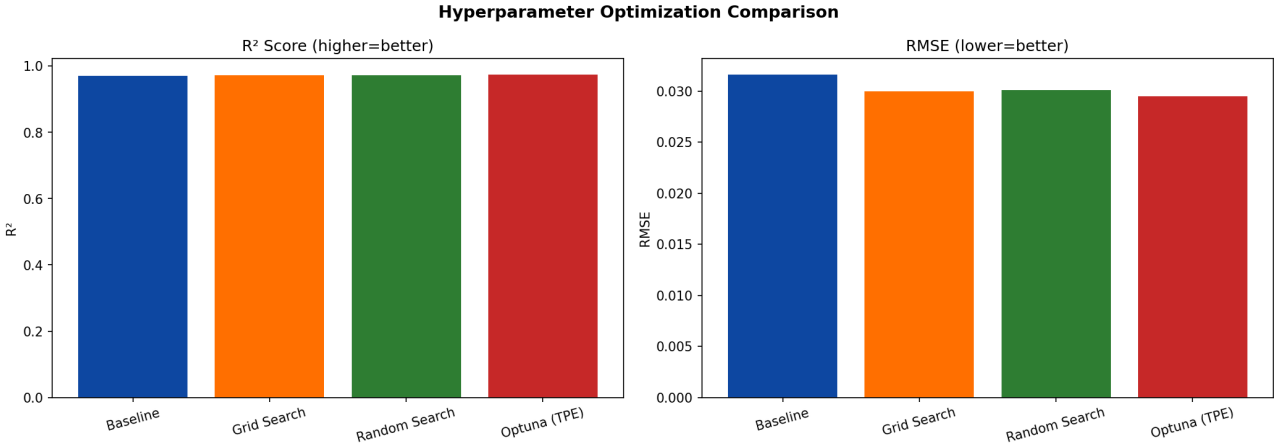

Hyperparameter Search Space Visualization
(Why exhaustive grid search is expensive in high dimensions)
3D Hyperparameter Landscape
(3D Printer Quality)
n_estimators
max_depth
R² Score
Top-Down View: Contour Map
(Darker = higher R²)
Optimal
max_depth
n_estimators

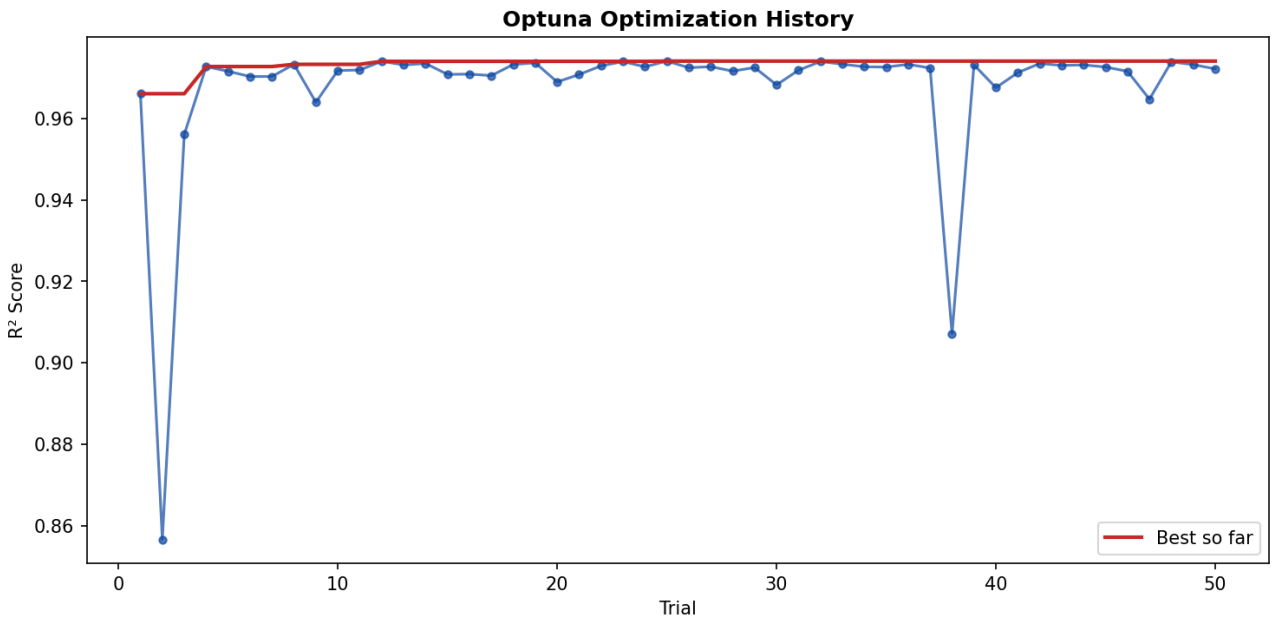

Grid Search vs Random Search vs Bayesian Optimization
(Same 30 evaluations — Bayesian finds better solutions faster)
Grid Search: Exhaustive
18 evaluations | Best R² = 0.935
param_max_depth
param_n_estimators
Random Search: 30 random trials
Best R² = 0.951
Best = 0.951
R² Score
Trial
Bayesian (Optuna): 30 intelligent trials
Best R² = 0.954
Best so far
Trial score
R² Score
Trial

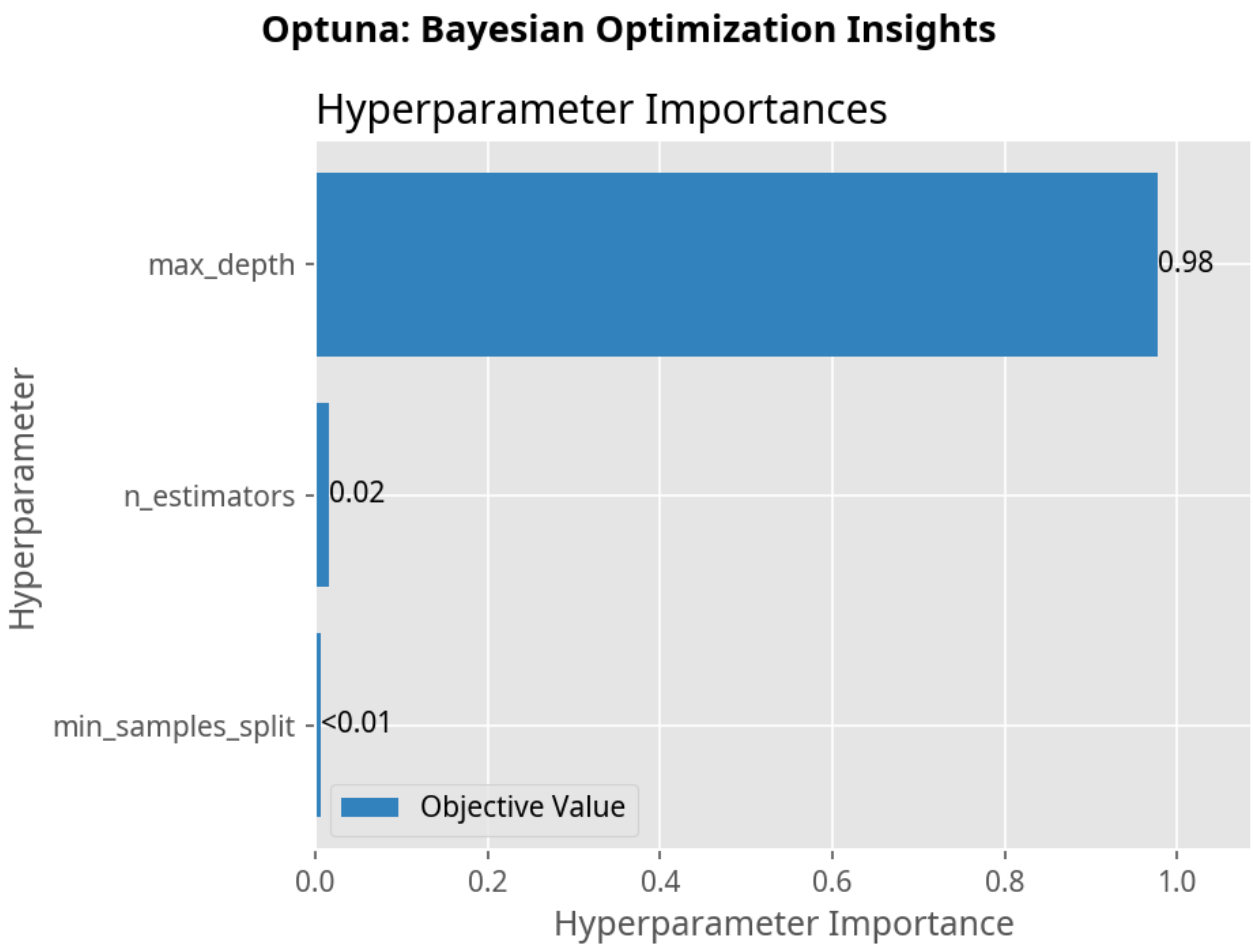

Bayesian Optimization for PLL Loop Filter Tuning
(Post-Silicon Validation: Minimizing Lock Time)
Bayesian Optimization: PLL Lock Time
Best = 2.6 μs
Best lock time
Trial
Lock Time (μs)
Trial
Optimal PLL Parameters Found
(Minimizes lock time to 2.6 μs)
ref_freq_mhz
60.4
vco_gain_mhz_v
508.8
phase_margin_deg
55.0
loop_bw_khz
386.6
charge_pump_ua
205.0
Parameter Value

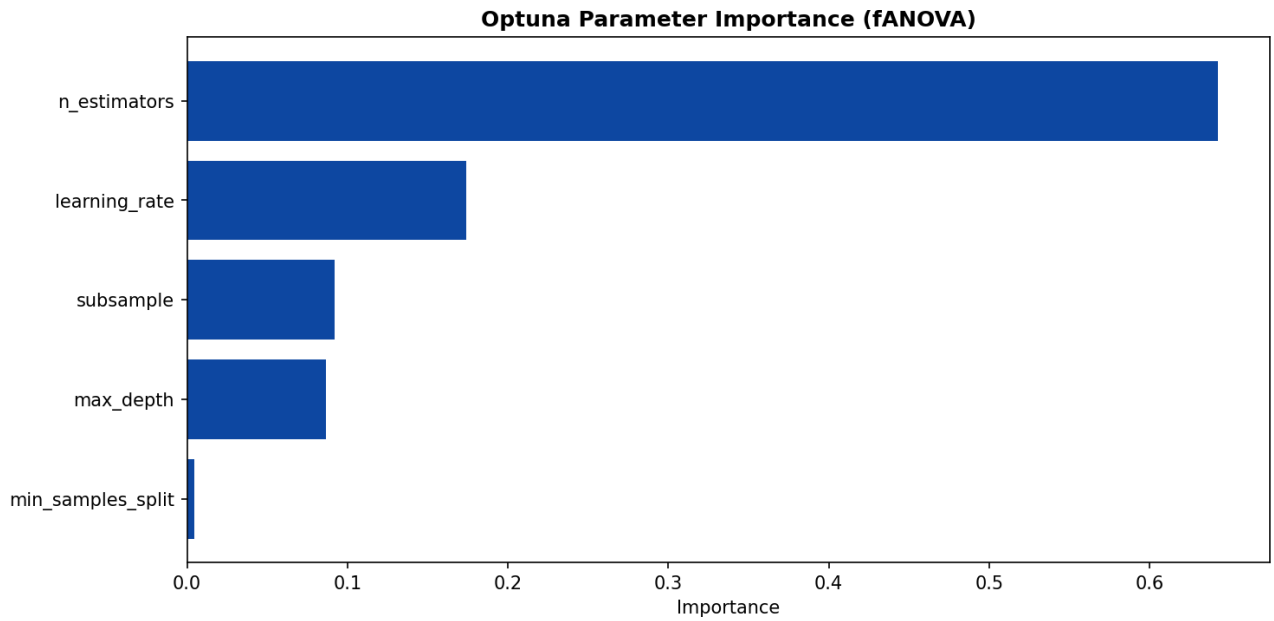

Cross-Validation: The Right Way to Evaluate Hyperparameter Tuning
5-Fold Cross-Validation
(Blue = validation, Gray = training)
Fold 5
Fold 4
Fold 3
Fold 2
Fold 1
Validation fold
Training data
0
20
40
60
80
100
Sample Index
Cross-Validation Scores: Grid vs Bayesian
(Consistent performance across folds = no overfitting)
Grid Search (mean=0.935)
Bayesian (mean=0.955)
R² Score
1.0
0.9
0.8
0.7
0.6
0.5
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5


PLL Loop Filter HPO Results
Optimization History
R²
0.96
0.94
0.92
0.90
0.88
0.86
0.84
0.82
Best
0
10
20
30
40
50
Trial
Method Comparison
R²
1.0
0.8
0.6
0.4
0.2
0.0
Baseline
Grid
Optuna


Hyperparameter Optimization Comparison
R² Score (higher=better)
R²
1.0
0.8
0.6
0.4
0.2
0.0
Baseline
Grid Search
Random Search
Optuna (TPE)
RMSE (lower=better)
RMSE
0.030
0.025
0.020
0.015
0.010
0.005
0.000
Baseline
Grid Search
Random Search
Optuna (TPE)


Optuna Optimization History
R² Score
0.96
0.94
0.92
0.90
0.88
0.86
0
10
20
30
40
50
Trial
Best so far


Optuna: Bayesian Optimization Insights
Hyperparameter Importances
Hyperparameter
max_depth
n_estimators
min_samples_split
0.98
0.02
<0.01
Objective Value
0.0
0.2
0.4
0.6
0.8
1.0
Hyperparameter Importance



The visualizations illustrate the convergence rates of each optimization method and parameter significance across datasets.

5 Tech Stack & Key Learnings

Info	Detail
GitHub	`https://github.com/AIML-Engineering-Lab/010_hyperparameter_optimization`
Tech Stack	Python 3.11, scikit-learn, pandas, Matplotlib
Licence	MIT

- Grid Search is effective for small spaces.
- Random Search is faster and simpler for medium spaces.
- Bayesian Optimization significantly optimizes performance in larger search spaces.